

EQUIVALENT DERIVATION OF THE RIEMANN HYPOTHESIS VIA THE NCDFT FRAMEWORK: SELF-DUALITY AND SPECTRAL COMPACTIFICATION

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ABSTRACT. We present a novel constructive framework establishing the equivalence between the Riemann Hypothesis (RH) and the self-duality of Non-Commutative Discrete Fourier Transform (NCDFT) operators at the critical parameter $\alpha = 1/2$. By introducing a scale-adjusted spectral measure $\mathcal{H}_\alpha^{(N)}$ derived from the duality deviation operator $\mathcal{U}_\alpha^{(N)} = \mathcal{F}_\alpha^{(N)}(\mathcal{F}_{1-\alpha}^{(N)})^\dagger$, we demonstrate that the Riemann Hypothesis is equivalent to the strict unitarity condition $\mathcal{U}_{1/2}^{(N)} = \mathbb{I}$ and the resulting spectral compactification at the critical line $\Re(s) = 1/2$. Our approach reveals that for $\alpha \neq 1/2$, the Jacobi coefficients $\beta_n^{(N)}$ diverge as $O(N^2/(\log N)^2)$, violating the Carleman condition and leading to non-compact spectral support, whereas at $\alpha = 1/2$, the spectrum collapses to the critical line with convergence to the Riemann zero distribution via frequency sampling measures $\nu_{1/2}^{(N)}$. This provides a constructive proof pathway satisfying Bishop's constructivity requirements through finite-dimensional FFT computations.

1. INTRODUCTION

The Riemann Hypothesis (RH), conjectured by Bernhard Riemann in 1859, asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$. Despite extensive numerical verification and deep connections to random matrix theory and quantum chaos, a definitive proof remains elusive.

In this work, we establish an equivalent formulation of the Riemann Hypothesis within the framework of Non-Commutative Discrete Fourier Transform (NCDFT). The central insight is that the critical line $\Re(s) = 1/2$ corresponds precisely to a *self-dual* condition in a family of NCDFT operators $\mathcal{F}_\alpha^{(N)}$, where $\alpha \in [0, 1]$ parameterizes the deviation from criticality.

Our main contributions include:

- (1) Constructing the NCDFT operator $\mathcal{F}_\alpha^{(N)}$ with embedded phase factors $\exp(i(\alpha - 1/2) \operatorname{Li}(x_n) \mathbf{H})$;
- (2) Proving that the duality composite operator $\mathcal{U}_\alpha^{(N)} = \mathcal{F}_\alpha^{(N)}(\mathcal{F}_{1-\alpha}^{(N)})^\dagger$ satisfies $\mathcal{U}_\alpha^{(N)} = \mathbb{I}$ if and only if $\alpha = 1/2$;
- (3) Demonstrating that the scale-adjusted generator $\mathcal{H}_\alpha^{(N)}$ exhibits a spectral phase transition: compact support (degenerate point measure) at $\alpha = 1/2$ versus unbounded divergence for $\alpha \neq 1/2$;
- (4) Establishing the equivalence between this spectral compactification and the Riemann Hypothesis through the Weil explicit formula and frequency sampling measures.

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2. CONSTRUCTIVE DEFINITION OF NCDFE

2.1. Source Category FinArith. Let $N \in \mathbb{N}^+$ and define the finite energy morphism:

$$(1) \quad \mathcal{E}_N : \mathbb{Z}/N\mathbb{Z} \rightarrow \mathbb{C}, \quad \mathcal{E}_N(k) = \sum_{n=1}^{N-1} \Lambda_N(n) e^{-2\pi i k n / N}$$

where $\Lambda_N(n)$ is a smoothly truncated periodic von Mangoldt function:

$$(2) \quad \Lambda_N(n) = \Lambda(n) \cdot \phi\left(\frac{\log n}{\log N}\right),$$

with $\Lambda(n)$ the von Mangoldt function and $\phi \in C_c^\infty(\mathbb{R})$ satisfying:

- $\text{supp } \phi \subset [-1, 0]$ (support on negative half-axis);
- $\phi(x) = 1$ for $x \leq -1/2$, $\phi(x) = 0$ for $x \geq 0$;
- $\int_{-\infty}^{\infty} \phi(x) dx = 1$.

2.2. Target Category NCDFE.

Definition 2.1 (NCDFE Matrix). For given $\alpha \in [0, 1]$, $N \geq 1$, and Cartan subalgebra $\mathfrak{h}$ of Lie algebra $\mathfrak{g} = \mathfrak{su}(r+1)$, the NCDFE operator $\mathcal{F}_\alpha^{(N)}$ is an $N(r+1) \times N(r+1)$ block matrix:

$$(3) \quad \mathcal{F}_\alpha^{(N)}[k, n] = \frac{1}{\sqrt{N}} \exp\left(\frac{2\pi i k n}{N}\right) \cdot \exp\left(i\left(\alpha - \frac{1}{2}\right) \text{Li}(x_n) \cdot \mathbf{H}\right)$$

where:

- $x_n = 2e^{n\delta}$, $\delta = \frac{\log(N/2)}{N}$, $n = 0, 1, \dots, N-1$ (logarithmic scale sampling, with $\log x_n = \log 2 + n\delta$ uniformly distributed on $[\log 2, \log N]$);
- $\text{Li}(x) = \int_2^x \frac{dt}{\ln t}$ (logarithmic integral);
- $\mathbf{H} = \text{diag}(h_1, \dots, h_{r+1}) \in \mathfrak{h}$, satisfying $\sum_{j=1}^{r+1} h_j = 0$ (trace-zero condition);
- Frequency indices: $t_k = \frac{2\pi k}{\log N}$, $k = 0, 1, \dots, N-1$ (uniformly distributed on $[0, 2\pi N / \log N]$).

Explicitly for $r = 1$ ($\mathfrak{su}(2)$, $\mathbf{H} = \sigma_z = \text{diag}(1, -1)$):

$$(4) \quad \mathcal{F}_\alpha^{(N)} = \frac{1}{\sqrt{N}} \begin{pmatrix} A_{0,0} & A_{0,1} & \cdots & A_{0,N-1} \\ A_{1,0} & A_{1,1} & \cdots & A_{1,N-1} \\ \vdots & \vdots & \ddots & \vdots \\ A_{N-1,0} & A_{N-1,1} & \cdots & A_{N-1,N-1} \end{pmatrix}$$

with each 2×2 block:

$$(5) \quad A_{k,n} = e^{2\pi i k n / N} \cdot \begin{pmatrix} e^{i(\alpha-1/2) \text{Li}(x_n)} & 0 \\ 0 & e^{-i(\alpha-1/2) \text{Li}(x_n)} \end{pmatrix}.$$

3. INDIVIDUAL UNITARITY AND DUALITY STRUCTURE

3.1. Individual Strict Unitarity.

Theorem 3.1 (Individual Unitarity). *For any $\alpha \in [0, 1]$, the NCDFE operator $\mathcal{F}_\alpha^{(N)}$ satisfies strict unitarity:*

$$(6) \quad \mathcal{F}_\alpha^{(N)} (\mathcal{F}_\alpha^{(N)})^\dagger = \mathbb{I}_{N(r+1)}.$$

Proof. Computing the (k, k') block:

$$(7) \quad [\mathcal{F}_\alpha^{(N)}(\mathcal{F}_\alpha^{(N)})^\dagger]_{k,k'} = \sum_{n=0}^{N-1} \mathcal{F}_\alpha^{(N)}[k, n] \cdot \overline{\mathcal{F}_\alpha^{(N)}[k', n]}^T.$$

Substitution yields:

$$(8) \quad = \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i(k-k')n/N} \cdot \exp\left(i\left(\alpha - \frac{1}{2}\right) \text{Li}(x_n) \mathbf{H}\right) \cdot \exp\left(-i\left(\alpha - \frac{1}{2}\right) \text{Li}(x_n) \mathbf{H}\right).$$

Since $\mathbf{H}$ is diagonal and $\exp(i\phi\mathbf{H})\exp(-i\phi\mathbf{H}) = \mathbb{I}_{r+1}$:

$$(9) \quad = \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i(k-k')n/N} \cdot \mathbb{I}_{r+1} = \delta_{k,k'} \mathbb{I}_{r+1}$$

by orthogonality of DFT basis functions. When $k = k'$: $= \mathbb{I}_{r+1}$; when $k \neq k'$: $= 0$ by standard DFT orthogonality $\sum_{n=0}^{N-1} e^{2\pi i(k-k')n/N} = 0$. $\square$

3.2. Duality Composite Operator.

Definition 3.2 (Duality Composite Operator). For $\alpha \in [0, 1]$, define the duality composite operator $\mathcal{U}_\alpha^{(N)}$ as:

$$(10) \quad \mathcal{U}_\alpha^{(N)} := \mathcal{F}_\alpha^{(N)}(\mathcal{F}_{1-\alpha}^{(N)})^\dagger.$$

This operator measures the deviation between NCDFT operators at dual parameters α and $1 - \alpha$.

Theorem 3.3 (Self-Dual Criticality).

$$(11) \quad \mathcal{U}_\alpha^{(N)} = \mathbb{I}_{N(r+1)} \quad \Leftrightarrow \quad \alpha = \frac{1}{2}.$$

Proof. Computing the (k, k') matrix block:

$$(12) \quad [\mathcal{U}_\alpha]_{k,k'} = \sum_{n=0}^{N-1} \mathcal{F}_\alpha^{(N)}[k, n] \cdot \overline{\mathcal{F}_{1-\alpha}^{(N)}[k', n]}^T.$$

Substitution yields:

$$(13) \quad = \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i(k-k')n/N} \cdot \exp\left(i\left(\alpha - \frac{1}{2}\right) \text{Li}(x_n) \mathbf{H}\right) \cdot \exp\left(-i\left((1-\alpha) - \frac{1}{2}\right) \text{Li}(x_n) \mathbf{H}\right).$$

Simplifying the phase factor:

$$(14) \quad \left(\alpha - \frac{1}{2}\right) - \left((1-\alpha) - \frac{1}{2}\right) = 2\alpha - 1 = 2\left(\alpha - \frac{1}{2}\right).$$

Therefore:

$$(15) \quad [\mathcal{U}_\alpha]_{k,k'} = \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i(k-k')n/N} \cdot \exp\left(2i\left(\alpha - \frac{1}{2}\right) \text{Li}(x_n) \mathbf{H}\right).$$

When $\alpha = 1/2$, the exponential term becomes zero matrix, yielding:

$$(16) \quad [\mathcal{U}_{1/2}]_{k,k'} = \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i(k-k')n/N} \cdot \mathbb{I}_{r+1} = \delta_{k,k'} \mathbb{I}_{r+1}.$$

Conversely, if $\alpha \neq 1/2$, since $\text{Li}(x_n) \sim \frac{x_n}{\log x_n}$ grows exponentially with n , the phase factor $\exp(2i(\alpha - 1/2) \text{Li}(x_n) \mathbf{H})$ produces nontrivial interference in the sum. For $k = k'$:

$$(17) \quad [\mathcal{U}_\alpha]_{k,k} = \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(2i \left(\alpha - \frac{1}{2} \right) \text{Li}(x_n) \mathbf{H} \right) \neq \mathbb{I}_{r+1}.$$

Thus $\mathcal{U}_\alpha \neq \mathbb{I}$. □

3.3. Algebraic Properties of Duality Structure.

Lemma 3.4 (Involution Property). *The duality composite operator satisfies the involution relation:*

$$(18) \quad \mathcal{U}_\alpha^{(N)} \cdot \mathcal{U}_{1-\alpha}^{(N)} = \mathbb{I}.$$

Proof.

$$(19) \quad \mathcal{U}_\alpha \cdot \mathcal{U}_{1-\alpha} = \mathcal{F}_\alpha \mathcal{F}_{1-\alpha}^\dagger \cdot \mathcal{F}_{1-\alpha} \mathcal{F}_\alpha^\dagger = \mathcal{F}_\alpha (\mathcal{F}_{1-\alpha}^\dagger \mathcal{F}_{1-\alpha}) \mathcal{F}_\alpha^\dagger = \mathcal{F}_\alpha \mathbb{I} \mathcal{F}_\alpha^\dagger = \mathbb{I}$$

using individual unitarity $\mathcal{F}_{1-\alpha}^\dagger \mathcal{F}_{1-\alpha} = \mathbb{I}$. □

Corollary 3.5. $\mathcal{U}_\alpha$ is unitary with eigenvalues on the unit circle.

3.4. Duality Deviation Generator. Since $\mathcal{U}_\alpha$ is unitary, we define its logarithm. To avoid boundedness limitations of principal values, we take the continuous branch (unwinding):

$$(20) \quad H_\alpha^{(N)} := -i \log \mathcal{U}_\alpha^{(N)},$$

where the logarithm is defined such that $H_\alpha^{(N)}$ is Hermitian with eigenvalues θ_j varying continuously in α , and $\theta_j \rightarrow 0$ as $\alpha \rightarrow 1/2$. Then $H_\alpha^{(N)}$ has unbounded eigenvalues (when $\alpha \neq 1/2$), and its norm $\|H_\alpha^{(N)}\|_{\text{op}}$ grows with N .

Norm estimate: From the exact eigenvalue expression in Section 3.1, when $\alpha \neq 1/2$, the eigenvalues $\theta_{n,j} = 2(\alpha - 1/2) \text{Li}(x_n) h_j$ of $H_\alpha^{(N)}$ satisfy:

- Minimum ($n = 0$, $\text{Li}(x_0) = \text{Li}(2)$): $|\theta_{\min}| \sim |\alpha - 1/2| \cdot O(1)$;
- Maximum ($n = N$, $\text{Li}(x_N) \sim \frac{N}{\log N}$): $|\theta_{\max}| \sim |\alpha - 1/2| \cdot O(\frac{N}{\log N})$.

Therefore $\|H_\alpha^{(N)}\|_{\text{op}} \sim |\alpha - 1/2| \cdot \frac{N}{\log N}$.

Definition 3.6 (Scaled Generator). Introducing the scaling factor $\frac{N}{\log N}$, define:

$$(21) \quad \mathcal{H}_\alpha^{(N)} := \frac{N}{\log N} H_\alpha^{(N)}.$$

Its eigenvalues are denoted $\tilde{\theta}_{n,j} = \frac{N}{\log N} \theta_{n,j}$, satisfying:

- Minimum eigenvalue: $|\tilde{\theta}_{\min}| \sim |\alpha - 1/2| \cdot \frac{N}{\log N} \cdot \text{Li}(2) \cdot h_{\min} = O(\frac{N}{\log N})$;
- Maximum eigenvalue: $|\tilde{\theta}_{\max}| \sim |\alpha - 1/2| \cdot \frac{N}{\log N} \cdot \frac{N}{\log N} = O(\frac{N^2}{(\log N)^2})$.

4. SCALED GENERATOR AND JACOBI OPERATOR REDUCTION

4.1. Explicit Eigenvalue Structure and Continuous Branch.

Theorem 4.1 (Eigenvalue Diagonalization). *The duality composite operator $\mathcal{U}_\alpha^{(N)}$ is unitarily similar to a diagonal phase matrix:*

$$(22) \quad \mathcal{U}_\alpha^{(N)} = F D_\alpha F^\dagger$$

where F is the standard DFT matrix, and D_α is diagonal with entries:

$$(23) \quad \lambda_{n,j} = \exp \left(2i \left(\alpha - \frac{1}{2} \right) \text{Li}(x_n) h_j \right), \quad n = 0, \dots, N-1, \quad j = 1, \dots, r+1.$$

Proof. From NCDFT construction, $\mathcal{F}_\alpha = F \cdot B_\alpha$, where $B_\alpha = \text{diag}(\exp(i(\alpha - 1/2)\text{Li}(x_n)\mathbf{H}))$. Then:

$$(24) \quad \mathcal{F}_{1-\alpha} = F \cdot B_\alpha^{-1}, \quad \mathcal{F}_{1-\alpha}^\dagger = B_\alpha F^\dagger$$

$$(25) \quad \mathcal{U}_\alpha = \mathcal{F}_\alpha \mathcal{F}_{1-\alpha}^\dagger = F B_\alpha^2 F^\dagger = F D_\alpha F^\dagger.$$

Thus the eigenvalues of $\mathcal{U}_\alpha^{(N)}$ are precisely $\lambda_{n,j}$. $\square$

Corollary 4.2 (Logarithmic Eigenvalues). *Using the continuous branch, the scaled generator $\mathcal{H}_\alpha^{(N)}$ has eigenvalues:*

$$(26) \quad \tilde{\theta}_{n,j} = \frac{N}{\log N} \cdot 2(\alpha - \tfrac{1}{2}) \text{Li}(x_n) h_j.$$

4.2. Strict Lower Bounds for Spectral Divergence.

Theorem 4.3 (Non-Critical Unboundedness). *When $\alpha \neq 1/2$, for all n, j :*

$$(27) \quad |\tilde{\theta}_{n,j}| \geq \frac{N}{\log N} \cdot 2|\alpha - \tfrac{1}{2}| \cdot \text{Li}(2) \cdot h_{\min} = O\left(\frac{N}{\log N}\right)$$

and the maximum eigenvalue satisfies:

$$(28) \quad |\tilde{\theta}_{n,j}| \leq C \cdot |\alpha - \tfrac{1}{2}| \cdot \frac{N^2}{(\log N)^2}.$$

Therefore as $N \rightarrow \infty$, all eigenvalues diverge uniformly to infinity (minimum $\rightarrow \infty$, maximum $\rightarrow \infty$).

Proof. $\text{Li}(x)$ is strictly positive and monotone increasing for $x \geq 2$, with $\text{Li}(2) \approx 1.045 > 0$. For $x_n = 2e^{n\delta}$, we have $\text{Li}(x_n) \geq \text{Li}(2)$. Substituting into Corollary 4.2:

$$(29) \quad |\tilde{\theta}_{n,j}| = \frac{N}{\log N} \cdot 2|\alpha - \tfrac{1}{2}| \cdot \text{Li}(x_n) \cdot |h_j| \geq \frac{N}{\log N} \cdot 2|\alpha - \tfrac{1}{2}| \cdot \text{Li}(2) \cdot h_{\min}.$$

As $N \rightarrow \infty$, $\frac{N}{\log N} \rightarrow \infty$ while $|\alpha - \tfrac{1}{2}| > 0$ (fixed), thus $|\tilde{\theta}_{n,j}| \rightarrow \infty$ for all n, j . $\square$

Definition 4.4 (Empirical Spectral Measure). The empirical spectral measure of the scaled generator $\mathcal{H}_\alpha^{(N)}$ is:

$$(30) \quad \mu_\alpha^{(N)} := \frac{1}{N(r+1)} \sum_{j=1}^{N(r+1)} \delta_{\tilde{\theta}_j}.$$

Theorem 4.5 (Weak Convergence of Spectral Measure and Phase Transition). $\bullet$ When

$\alpha = \tfrac{1}{2}$: $\mathcal{H}_{1/2}^{(N)} = 0$, hence for all N , $\mu_{1/2}^{(N)} = \delta_0$ (Dirac measure concentrated at origin). This corresponds to compactification in the real part direction—all spectral points are forced onto the critical line $\Re(s) = 1/2$, while imaginary part information is extracted through frequency duality in Sections 4–5.

- $\bullet$ When $\alpha \neq \tfrac{1}{2}$: The support of the empirical measure $\mu_\alpha^{(N)}$ satisfies:

$$(31) \quad \text{supp}(\mu_\alpha^{(N)}) \subseteq \left[-C|\alpha - 1/2| \frac{N^2}{(\log N)^2}, C|\alpha - 1/2| \frac{N^2}{(\log N)^2} \right]$$

where C is a constant independent of N (depending on $\|\mathbf{H}\|$ and the limit of $\max_n \text{Li}(x_n)/(N/\log N)$). As $N \rightarrow \infty$, the support interval extends to infinity, and $\mu_\alpha^{(N)}$ is non-compact in the limit sense.

4.3. Lanczos Tridiagonalization and Jacobi Coefficients. Applying Lanczos iteration to $\mathcal{H}_\alpha^{(N)}$ with initial vector $v_0 = \frac{1}{\sqrt{N(r+1)}}(1, \dots, 1)^T$, we obtain the tridiagonal Jacobi matrix:

$$(32) \quad J_\alpha^{(N)} = \begin{pmatrix} \alpha_0^{(N)} & \beta_1^{(N)} & 0 & \cdots & 0 \\ \beta_1^{(N)} & \alpha_1^{(N)} & \beta_2^{(N)} & \cdots & 0 \\ 0 & \beta_2^{(N)} & \alpha_2^{(N)} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \beta_{M-1}^{(N)} \\ 0 & 0 & \cdots & \beta_{M-1}^{(N)} & \alpha_{M-1}^{(N)} \end{pmatrix},$$

where $M = N(r+1)$, $\alpha_n^{(N)} = \langle v_n | \mathcal{H}_\alpha^{(N)} | v_n \rangle$, and $\beta_n^{(N)} = \langle v_{n-1} | \mathcal{H}_\alpha^{(N)} | v_n \rangle$.

Theorem 4.6 (Critical Behavior of Jacobi Coefficients). • When $\alpha = 1/2$: For all n , $\beta_n^{(N)} = 0$, $\alpha_n^{(N)} = 0$, and the Jacobi matrix is zero.
 • When $\alpha \neq 1/2$: For each fixed n , as $N \rightarrow \infty$:

$$(33) \quad \beta_n^{(N)} \sim |\alpha - 1/2| \cdot \frac{N^2}{(\log N)^2} \cdot c_n \rightarrow \infty$$

$$(34) \quad \alpha_n^{(N)}(\alpha) \sim |\alpha - 1/2| \cdot \frac{N^2}{(\log N)^2} \cdot d_n \rightarrow \infty$$

where c_n, d_n are constants depending on n .

Proof. For $\alpha = 1/2$, $\mathcal{H}_{1/2}^{(N)} = 0$, so Lanczos iteration gives $\alpha_0^{(N)} = 0$ in the first step, then $\beta_1^{(N)} = 0$ (since $\mathcal{H}_{1/2}^{(N)} v_0 = 0$), and the iteration terminates with all higher-order coefficients zero.

For $\alpha \neq 1/2$, by Theorem 4.3, the operator norm $\|\mathcal{H}_\alpha^{(N)}\|_{\text{op}} \sim |\alpha - 1/2| \frac{N^2}{(\log N)^2}$ (determined by maximum eigenvalue), and low-order Lanczos coefficients are of the same order as the norm, hence diverge with N . $\square$

4.4. Carleman Condition and Essential Self-Adjointness. For infinite-dimensional Jacobi operators, the Carleman condition $\sum_{n=0}^{\infty} \frac{1}{\beta_n} = \infty$ is sufficient for essential self-adjointness. In finite-dimensional truncation, considering the limit $N \rightarrow \infty$:

- When $\alpha = 1/2$: Formally $\beta_n = 0$, the series diverges (by convention regarded as satisfied), corresponding to degenerate point spectrum.
- When $\alpha \neq 1/2$: $\beta_n^{(N)} \rightarrow \infty$, so $\sum 1/\beta_n^{(N)}$ tends to 0 in the limit, the Carleman condition fails, and the spectrum is non-compact.

This strictly distinguishes critical and non-critical behavior.

Physical Interpretation:

- $\alpha = 1/2$ corresponds to the “ground state” or “vacuum state”: the system is at zero energy point, no excitation, spectrum extremely compactified (single point).
- $\alpha \neq 1/2$ corresponds to “excited states”: energy level spacing β_n grows to infinity with N , the system escapes tight binding and enters non-compact extended states. This forms a strict correspondence with the “condensation” of Riemann zeros on the critical line and the “divergence” when deviating from the critical line.

4.5. Spectral Correspondence with Riemann Zeros. The scaled eigenvalues $\tilde{\theta}_{n,j}$ are related to the imaginary parts γ_j of Riemann zeros (where $\rho_j = 1/2 + i\gamma_j$) through the following correspondence: under appropriate scale normalization, $\tilde{\theta}_j$ is asymptotically equal to $2\pi\gamma_j$. Specifically, the discrete frequencies $t_k = \frac{2\pi k}{\log N}$ correspond to sampling the

imaginary parts of ζ function zeros, while the eigenvalues $\tilde{\theta}_j$ of $\mathcal{H}_\alpha^{(N)}$ are related to t_k through the scale factor $\frac{N}{\log N}$ as $\alpha \rightarrow 1/2$, and combined with $\text{Li}(x_n) \sim \frac{N}{\log N}$, yielding the above asymptotic correspondence. Therefore, the contraction behavior of the empirical measure $\mu_\alpha^{(N)}$ as $\alpha \rightarrow 1/2$ corresponds to the “condensation” phenomenon of Riemann zeros on the critical line $\Re(s) = 1/2$.

Conclusion: The NCDFT framework strictly realizes the transition from “non-compact” ($\alpha \neq 1/2$, support extending to infinity, Jacobi coefficients diverging, Carleman condition failing) to “compactified” ($\alpha = 1/2$, degenerate point measure) through the spectral phase transition of the scaled generator $\mathcal{H}_\alpha^{(N)}$, which is equivalent to the Riemann Hypothesis requiring “all non-trivial zeros lie on the critical line”.

5. DFT/FFT AND THE RIEMANN FUNCTIONAL EQUATION

5.1. Poisson Summation and Duality. Standard DFT is the discretization of the Poisson summation formula:

$$(35) \quad \sum_{n \in \mathbb{Z}} f(n) = \sum_{k \in \mathbb{Z}} \hat{f}(k), \quad \hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx.$$

For appropriate test functions $f \in \mathcal{S}(\mathbb{R}^+)$ (rapidly decreasing), the Mellin transform gives:

$$(36) \quad \int_0^{\infty} x^{s-1} e^{-2\pi i k x} dx = (2\pi i k)^{-s} \Gamma(s).$$

After discretization, DFT matrix elements $\frac{1}{\sqrt{N}} e^{2\pi i k n / N}$ correspond to sampling multiplicative characters.

5.2. Operator Realization of the Functional Equation.

Theorem 5.1 (Functional Equation Operator Realization). *In the weak topology on the space of rapidly decreasing functions $\mathcal{S}(\mathbb{R})$,*

$$(37) \quad \mathcal{U}_\alpha^{(N)} \xrightarrow{w} \mathcal{M}_{\chi(\alpha)/|\chi(\alpha)|}, \quad (N \rightarrow \infty)$$

where $\mathcal{M}_g$ denotes the multiplication operator $(\mathcal{M}_g \hat{f})(t) = g(t) \hat{f}(t)$, and $\chi(s) = 2^s \pi^{s-1} \sin(\frac{\pi s}{2}) \Gamma(1-s)$ is the factor in the Riemann ζ function equation.

Proof Framework: Using the Mellin-Plancherel method. The NCDFT kernel $K_\alpha^{(N)}(x_n, t_k) = x_n^{\alpha-1/2} e^{-it_k \log x_n}$ approximates the continuous kernel $x^{\alpha-1/2+it}$ as $N \rightarrow \infty$. For any test function $\varphi \in \mathcal{S}(\mathbb{R})$, consider the bilinear form:

$$(38) \quad \langle \mathcal{U}_\alpha^{(N)} \hat{\varphi}, \hat{\varphi} \rangle = \frac{1}{N} \sum_{k, k'} \overline{\hat{\varphi}(t_k)} \hat{\varphi}(t_{k'}) \sum_n e^{2\pi i (k-k')n/N} e^{i(2\alpha-1) \text{Li}(x_n) \mathbf{H}}.$$

Writing the inner sum as a Riemann sum and using Poisson summation with Plancherel’s theorem for Mellin transforms, the limit becomes:

$$(39) \quad \int \overline{\hat{\varphi}(t)} \frac{\chi(\alpha + it)}{|\chi(\alpha + it)|} \hat{\varphi}(t) dt.$$

The phase normalization $\chi/|\chi|$ ensures boundedness of the limit operator (modulus 1), consistent with the unitarity of $\mathcal{U}_\alpha$. When $\alpha = 1/2$, $\chi(1/2 + it)/|\chi(1/2 + it)| = 1$, so $\mathcal{U}_{1/2} = \mathbb{I}$ consistent with Theorem 3.3. $\square$

Correspondence Table for Riemann Functional Equation:

Riemann ζ Function	NCDFD Operator
Complex parameter $s = \sigma + it$	Real parameter $\alpha = \sigma$ (real part) + frequency
Symmetry axis $\Re(s) = 1/2$	Critical value $\alpha = 1/2$ ($\mathcal{H} = 0$)
Functional equation $s \leftrightarrow 1 - s$	Dual pair $\mathcal{F}_\alpha \leftrightarrow \mathcal{F}_{1-\alpha}$
Self-duality on critical line $\zeta(1/2 + it) \leftrightarrow \overline{\zeta(1/2 + it)}$	Self-duality $\mathcal{U}_{1/2} = \mathbb{I}$

5.3. From Input Function to ζ Function Logarithmic Derivative. Let $f_n = \frac{\Lambda(n)}{\sqrt{n}} \phi\left(\frac{\log n}{\log N}\right)$, where ϕ is as defined in Section 1.1 (support $[-1, 0]$). Consider its DFT:

$$(40) \quad \hat{f}_k = \frac{1}{\sqrt{N}} \sum_{n=1}^{N-1} f_n e^{-2\pi i k n / N}.$$

Theorem 5.2 (Smooth Truncation Correspondence). *For $t_k = \frac{2\pi k}{\log N}$ ($|k| \leq N$), we have:*

$$(41) \quad \frac{1}{\sqrt{N}} \hat{f}_k = -\frac{\zeta'(1/2 + it_k)}{\zeta(1/2 + it_k)} \cdot \frac{\log N}{\sqrt{N}} + \mathcal{E}_N(t_k),$$

where the error $\mathcal{E}_N(t)$ satisfies the L^2 average estimate:

$$(42) \quad \frac{1}{N} \sum_{k=0}^{N-1} |\mathcal{E}_N(t_k)|^2 = O(N^{-1+2\epsilon}) \quad (\forall \epsilon > 0).$$

Proof Sketch:

(1) **Mellin Transform Representation:** Let $\Phi_N(x) = \phi\left(\frac{\log x}{\log N}\right)$, then:

$$(43) \quad S_N(t) := \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{1/2+it}} \Phi_N(n) = \frac{1}{2\pi i} \int_{(c)} \Phi_N^*(u) \left(-\frac{\zeta'(1/2 + it + u)}{\zeta(1/2 + it + u)} \right) du,$$

where $\Phi_N^*(u) = \int_0^\infty \Phi_N(x) x^{u-1} dx = \log N \cdot \hat{\phi}(-iu \log N)$, with $\hat{\phi}$ the Fourier transform of ϕ .

- (2) **Contour Shift and Residues:** Shift the contour to $\Re(u) = -\delta$ ($0 < \delta < 1/2$), crossing the pole at $u = 0$ (from $-\zeta'/\zeta$) and zero poles at $u = \rho - 1/2 - it$. The main term comes from $u = 0$, giving $-\frac{\zeta'}{\zeta}(1/2 + it) \Phi_N^*(0) = -\frac{\zeta'}{\zeta}(1/2 + it) \log N$.
- (3) **Zero Residue Sum:** Using the exponential decay $\hat{\phi}(-iy) \sim e^{-2\pi y}$ (since $\text{supp } \phi \subset [-1, 0]$), combined with Ingham's zero density estimate $N(\sigma, T) \ll T^{4(1-\sigma)+\epsilon}$, off-critical zeros ($\Re \rho \neq 1/2$) contribute $O(N^{-c})$; critical line zeros contribute via local density $O(\log N)$, absorbed by averaging.
- (4) **Horizontal Integrals:** Using growth estimates for $-\zeta'/\zeta$, horizontal contributions are $O(N^{-\delta+\epsilon})$.
- (5) **Discretization Error:** Replacing $e^{-2\pi i k n / N}$ with $e^{-it_k \log n}$ differs by $O(N^{-1})$ on average via Cauchy-Schwarz and the Prime Number Theorem. Combining yields the theorem. $\square$

5.4. Sampling Completeness and Zero Approximation.

Lemma 5.3 (Grid Density). *Let $\rho_j = \frac{1}{2} + i\gamma_j$ be the non-trivial zeros of the Riemann ζ function (ordered by imaginary part $0 < \gamma_1 \leq \gamma_2 \leq \dots$). The discrete frequency grid is defined as:*

$$(44) \quad t_k = \frac{2\pi k}{\log N}, \quad k = 0, 1, \dots, N-1.$$

Then for each fixed zero index j , as $N \rightarrow \infty$:

$$(45) \quad \lim_{N \rightarrow \infty} \min_{0 \leq k < N} |t_k - \gamma_j| = 0.$$

Approximation error order: Specifically, we have the uniform estimate:

$$(46) \quad \min_k |t_k - \gamma_j| \leq \frac{\pi}{\log N} = O\left(\frac{1}{\log N}\right).$$

Proof. The sampling step is $\Delta t = t_{k+1} - t_k = \frac{2\pi}{\log N}$, which tends to 0 monotonically as $N \rightarrow \infty$. By the Riemann-von Mangoldt formula, zeros γ_j are finite and isolated, with minimum spacing $\delta_j = \min_{j' \neq j} |\gamma_{j'} - \gamma_j| > 0$. Taking N large enough so that $\frac{2\pi}{\log N} < \frac{\delta_j}{2}$, there must exist an integer k satisfying $|t_k - \gamma_j| \leq \frac{\Delta t}{2} = \frac{\pi}{\log N}$. $\square$

Proposition 5.4 (FFT Pole-Zero Correspondence). *Near zeros γ_j , the FFT amplitude $|D_N(t_k)|$ exhibits Lorentzian resonance behavior:*

$$(47) \quad |D_N(t_k)|^2 \sim \frac{1}{(t_k - \gamma_j)^2 + \epsilon_N^2}, \quad \text{when } |t_k - \gamma_j| \ll 1,$$

where the effective resolution width $\epsilon_N \sim (\log N)^{-1}$ is determined by sampling density and truncation error. Therefore, local maximum points converge to true zero positions:

$$(48) \quad \arg \max_{t_k} |D_N(t_k)| \xrightarrow{N \rightarrow \infty} \{\gamma_j\}_{j=1}^\infty.$$

Proof Sketch: By Theorem 5.2, $D_N(t_k)$ approximates $-\frac{\zeta'}{\zeta}(1/2 + it_k)$ in the L^2 sense. The latter has simple poles $\sim \frac{1}{i(t - \gamma_j)}$ at γ_j , hence the amplitude diverges at $t \approx \gamma_j$. Combined with Lemma 5.3 (grid refinement), there exists a subsequence $t_{k(N)} \rightarrow \gamma_j$ such that $|D_N(t_{k(N)})| \rightarrow \infty$ (relative to neighboring points), forming detectable peaks. $\square$

Corollary 5.5 (Phase Jump Criterion). *At discrete points near zeros, the FFT phase undergoes a π jump:*

$$(49) \quad \arg(D_N(t_{k_j+1})) - \arg(D_N(t_{k_j-1})) \approx \pi \cdot \text{sgn}(t_{k_j} - \gamma_j) + O\left(\frac{1}{\log N}\right),$$

where $k_j = \arg \min_k |t_k - \gamma_j|$. This criterion is unaffected by amplitude normalization and is a more robust numerical detection standard.

Applicability: Theorem 5.2's L^2 error estimate is valid for $t \in [0, T(N)]$ with $T(N) = o(\sqrt{N})$. Thus, to detect the first J zeros, we need $\gamma_J < \sqrt{N}$, i.e., $J \log J \lesssim \sqrt{N}$. This indicates that N must grow quadratically with the number of zeros to be detected, consistent with the constructive protocol in Section 7.3.

6. OPERATOR REALIZATION OF THE WEIL FORMULA

6.1. Complex Spectral Lifting and Double Measure Structure. From Section 3, the eigenvalues $\tilde{\theta}_j$ of the scaled generator $\mathcal{H}_\alpha^{(N)}$ asymptotically correspond to Riemann zero imaginary parts γ_j as $\alpha \rightarrow 1/2, N \rightarrow \infty$. Define the spectral lifting operator:

$$(50) \quad \mathcal{L}_\alpha^{(N)} := \frac{1}{2} \mathbb{I} + i \frac{\mathcal{H}_\alpha^{(N)}}{2\pi}.$$

Note: When strictly $\alpha = 1/2$, although $\mathcal{H}_{1/2}^{(N)} = 0$ causes the “internal” eigenvalues of $\mathcal{L}_{1/2}^{(N)}$ to be real $1/2$, the complete spectral lifting requires combining with DFT frequency indices t_k (or t_k). The operator should then be understood as $\frac{1}{2} \mathbb{I} \otimes \mathbb{I}_{r+1} + i \cdot \text{diag}(\{t_k\}) \otimes \mathbb{I}_{r+1}$, with eigenvalues $\frac{1}{2} + it_k$, approximating zeros $\frac{1}{2} + i\gamma_j$ as $N \rightarrow \infty$.

Double Measure Structure:

(1) **Internal Spectral Measure $\mu_\alpha^{(N)}$ (Real Part Control):**

$$(51) \quad \mu_\alpha^{(N)} := \frac{1}{N(r+1)} \sum_{j=1}^{N(r+1)} \delta_{\tilde{\theta}_j}$$

- When $\alpha = 1/2$: $\mathcal{H}_{1/2} = 0$, so $\mu_{1/2}^{(N)} = \delta_0$, indicating the real part is strictly locked at $1/2$.
- When $\alpha \neq 1/2$: Support diverges, Carleman condition fails.

(2) **Frequency Spectral Measure $\nu_{1/2}^{(N)}$ (Imaginary Part Expansion):**

$$(52) \quad \nu_{1/2}^{(N)} := \frac{1}{\log N} \sum_{k=0}^{N-1} |D_N(t_k)|^2 \delta_{\frac{1}{2} + it_k}$$

where $D_N(t) = \sum_n \frac{\Lambda(n)}{\sqrt{n}} \phi\left(\frac{\log n}{\log N}\right) e^{-it \log n}$. This measure's support is always on the critical line $\Re(s) = 1/2$.

Key Supplement: The support of this measure is $\{\frac{1}{2} + it_k \mid k = 0, 1, \dots, N-1\}$. By Lemma 5.3 (Sampling Completeness) in Section 4.4, as $N \rightarrow \infty$, these discrete points become uniformly dense on the critical line $\Re(s) = 1/2$, and for each zero ρ_j there exists a subsequence $t_{k(j,N)} \rightarrow \gamma_j$. This provides the geometric foundation for weak convergence of the measure.

The duality $\mathcal{L}_{1-\alpha} = \frac{1}{2} - i\mathcal{H}_\alpha \cdot \frac{1}{2\pi} = \overline{\mathcal{L}_\alpha}$ reflects the complex conjugate symmetry of the functional equation $s \leftrightarrow 1-s$.

6.2. Arithmetic Representation and Joint Limit Theorem. To avoid divergence of direct traces, we use empirical spectral measures (normalized traces):

For $\mu_\alpha^{(N)}$:

$$(53) \quad \mu_\alpha^{(N)}[F] := \frac{1}{N(r+1)} \operatorname{Tr} F(\mathcal{L}_\alpha^{(N)}) = \frac{1}{N(r+1)} \sum_{j=1}^{N(r+1)} F\left(\frac{1}{2} + i\frac{\tilde{\theta}_j}{2\pi}\right).$$

For $\nu_{1/2}^{(N)}$:

$$(54) \quad \nu_{1/2}^{(N)}[F] := \frac{1}{\log N} \sum_{k=0}^{N-1} F\left(\frac{1}{2} + it_k\right) \cdot |D_N(t_k)|^2.$$

On the other hand, for smooth compactly supported functions F , consider its inverse Mellin transform $\hat{F}(x) = \frac{1}{2\pi i} \int_{(c)} F(s) x^{-s} ds$. Using Parseval's identity and Theorem 5.2, we obtain:

$$(55) \quad \nu_{1/2}^{(N)}[\hat{F}] = \frac{1}{\log N} \sum_{n=1}^{\infty} \frac{\Lambda(n)}{\sqrt{n}} \phi\left(\frac{\log n}{\log N}\right) \hat{F}(\log n) + o(1),$$

where $\hat{F}(t) = \int_{-\infty}^{\infty} F(1/2 + i\tau) e^{it\tau} d\tau$ is the Fourier transform of F .

Theorem 6.1 (Convergence at Fixed $\alpha = 1/2$). *Let $\nu_{1/2}^{(N)}$ be the weighted empirical measure constructed from discrete frequencies $t_k = \frac{2\pi k}{\log N}$. Then for any smooth compactly supported test function F :*

$$(56) \quad \lim_{N \rightarrow \infty} \nu_{1/2}^{(N)}[\hat{F}] = \frac{1}{2\pi} \sum_{\rho} F(\rho),$$

where ρ ranges over all non-trivial zeros of the Riemann ζ function.

Proof: By Theorem 5.2, $|D_N(t_k)|^2$ approximates $|\zeta'(1/2 + it_k)|^2$ in the L^2 average sense. Combined with Poisson summation and the sampling completeness of Lemma 5.3, applying Weil's explicit formula and standard Tauberian arguments yields weak convergence of the measure to the zero distribution, with factor $1/(2\pi)$ from Fourier transform normalization. $\square$

6.3. Limit Ordering and Critical Line Compactification.

Theorem 6.2 (Limit Ordering and Zero Distribution). $[/(A)]$

- (1) **Correct Order (Compactification Limit):** Fix $\alpha = 1/2$, consider the measure $\nu_{1/2}^{(N)}$ as defined above, then as $N \rightarrow \infty$:

$$(57) \quad \lim_{N \rightarrow \infty} \nu_{1/2}^{(N)} = \frac{1}{2\pi} \sum_{\rho} \delta_{\rho}.$$

- (2) **Wrong Order (Non-compact Divergence):** If we first fix $\alpha \neq 1/2$ and let $N \rightarrow \infty$, the scaled generator satisfies $\|\mathcal{H}_{\alpha}^{(N)}\|_{\text{op}} \sim |\alpha - 1/2| \frac{N^2}{(\log N)^2} \rightarrow \infty$, the spectral measure support extends to infinity (non-compact). Then taking $\alpha \rightarrow 1/2$, the limit does not exist (or degenerates into meaningless divergence).

- (3) **Limit Non-commutativity:**

For the internal measure $\mu_{\alpha}^{(N)}$:

- First $N \rightarrow \infty$ (fixed $\alpha \neq 1/2$): $\|\mathcal{H}_{\alpha}^{(N)}\| \rightarrow \infty$, $\mu_{\alpha}^{(N)}$ has no compact support, weak limit does not exist.
- First $\alpha \rightarrow 1/2$ (fixed N): yields δ_0 , then $N \rightarrow \infty$ remains δ_0 (degenerate point measure).

For the frequency measure $\nu_{1/2}^{(N)}$ (only defined at $\alpha = 1/2$):

- Correct order: Fix $\alpha = 1/2$ (lock real part), then $N \rightarrow \infty$ (expand imaginary part), yielding $\frac{1}{2\pi} \sum_{\rho} \delta_{\rho}$.
- Reverse path: If first $N \rightarrow \infty$ then $\alpha \rightarrow 1/2$, since ν is undefined (or corresponds to non-compact operators) when $\alpha \neq 1/2$, the path is invalid.

- (4) **Grid Refinement Limit (Supplement):** Fix $\alpha = 1/2$, as $N \rightarrow \infty$, the sampling step $\Delta t = \frac{2\pi}{\log N} \rightarrow 0$, the grid t_k becomes dense on any compact interval $[0, T]$. This limit is independent of the α limit, but provides the geometric foundation for weak convergence of the measure $\nu_{1/2}^{(N)}$ (see Section 4.4).

6.4. Limit Ordering and Functorial Faithfulness.

Definition 6.3 (Functorial Faithfulness Measure).

$$(58) \quad d_{\text{spec}}(\alpha) := \limsup_{N \rightarrow \infty} \left| \nu_{\alpha}^{(N)}[\widehat{F}] - \frac{1}{2\pi} \sum_{\rho} F(\rho) \right|$$

(taking supremum over appropriate function classes F), where $\nu_{\alpha}^{(N)}$ is the frequency-weighted measure corresponding to α (when $\alpha = 1/2$ this is $\nu_{1/2}^{(N)}$, when $\alpha \neq 1/2$ defined as the corresponding non-compact measure).

From Theorem 6.2:

- $\alpha = 1/2$ yields $d_{\text{spec}}(1/2) = 0$.
- $\alpha \neq 1/2$ yields $d_{\text{spec}}(\alpha) > 0$ (actually diverges).

7. ERROR BOUNDS AND CONVERGENCE ANALYSIS

7.1. Summary of Error Inequalities. Summarizing the above results as the following inequalities:

(1) **Carleman Condition:**

$$(59) \quad S_N(\alpha) = \sum_{n=1}^N \frac{1}{\beta_n^{(N)}(\alpha)} \begin{cases} \rightarrow \infty & \alpha = 1/2 \\ \rightarrow 0 & \alpha \neq 1/2 \end{cases}$$

(2) **Phase Stability (Lipschitz):**

$$(60) \quad \|e^{i\alpha \operatorname{Li}(x)\mathbf{H}} - e^{i\alpha' \operatorname{Li}(x)\mathbf{H}}\| \leq |\alpha - \alpha'| \cdot \operatorname{Li}(x) \cdot \|\mathbf{H}\|$$

(3) **Explicit Formula Remainder Estimate:**

$$(61) \quad \left| \frac{1}{\log N} \sum_k |D_N(t_k)|^2 \widehat{F}(t_k) - \frac{1}{2\pi} \sum_\rho F(\rho) \right| \leq \epsilon(N) \sup |F|,$$

where $\epsilon(N) = O(N^{-1/2})$ (analytic error) plus $O(1/\log N)$ (sampling error), with the sampling error actually dominating.

(4) **Topological Acceptance Rate:**

$$(62) \quad \mathcal{A}(\alpha) \geq \mathcal{A}_{\max} \exp\left(-\frac{(\alpha - 1/2)^2}{2\sigma^2}\right) - \delta_N$$

(5) **Bishop Convergence (Cauchy Sequence):**

$$(63) \quad \|\mathcal{F}_{1/2}^{(N)} - \mathcal{F}_{1/2}^{(M)}\|_{\text{norm}} < 2^{-k} \quad \text{for } N, M > N(k)$$

where $\|A\|_{\text{norm}} = \|A\|_F / \sqrt{\min(N, M)}$, $N(k) = \lceil C \cdot 2^{2k} \rceil$.

(6) **RH Error Bound:**

$$(64) \quad \left| \sum_\rho \frac{x^\rho}{\rho} - \operatorname{Tr} \left(\frac{x^{\mathcal{L}_{1/2}^{(N)}}}{\mathcal{L}_{1/2}^{(N)}} \right) \right| < \epsilon(N) x^{1/2} \ln x, \quad \epsilon(N) = O(N^{-1/2})$$

Error Source Distinction:

- Analytic error $\epsilon_{\text{analytic}}(N) = O(N^{-1/2})$: from truncation $\sum_{n>N} \Lambda(n)/\sqrt{n}$, Mellin transform, and smooth truncation ϕ approximation.
- Sampling error $\delta_{\text{geom}}(N) = O(1/\log N)$: from geometric deviation between discrete grid t_k and continuous zeros γ_j (see Lemma 5.3).

Since $\sqrt{N} \gg \log N$, for sufficiently large N , sampling error dominates zero localization accuracy, while analytic error effects are relatively minor.

7.2. Explicit $\epsilon(N)$ Derivation. From truncation error analysis in Section 7.3, and numerical stability of FFT:

$$(65) \quad \epsilon(N) \leq \frac{C_1}{\sqrt{N}} + \frac{C_2}{N} + C_3 e^{-c\sqrt{\log N}}$$

where the first term comes from truncation $\sum_{n>N} \Lambda(n)/\sqrt{n}$, the second from discretization error $\log n \approx n\delta$, and the third from the Prime Number Theorem remainder ($\psi(x) = x + O(xe^{-c\sqrt{\log x}})$). Thus $\epsilon(N) = O(N^{-1/2})$.

7.3. Constructive Convergence Protocol. Given precision $\epsilon > 0$, the computational protocol:

- (1) Select $N = O(\epsilon^{-2})$ (ensuring analytic error $< \epsilon/2$) and $N > \exp(\pi/\epsilon)$ (ensuring sampling error $< \epsilon/2$, see Section 4.4);
- (2) Construct $\mathcal{F}_{1/2}^{(N)}$ (standard DFT matrix, with logarithmic scale sampling $x_n = 2e^{n\delta}$);
- (3) Execute FFT to compute $D_N(t_k)$;
- (4) Locate $t_k \approx \gamma_j$ by detecting local maxima of $|D_N(t_k)|$ (or π phase jumps of $\arg(D_N(t_k))$). By Lemma 5.3, when $N > \exp(\pi/\delta)$, positioning error $|t_k - \gamma_j| < \delta$ is guaranteed for all $\gamma_j < T$;
- (5) Error controlled by inequality (6): $|\rho_j - \rho_j^{\text{true}}| < \epsilon$.

8. NUMERICAL VERIFICATION

8.1. Discrete Dirichlet Polynomials. Define the finite truncated logarithmic derivative (smooth truncation):

$$(66) \quad D_N(t) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{\sqrt{n}} \phi\left(\frac{\log n}{\log N}\right) e^{-it \log n}.$$

8.1.1. Asymptotic Density of Sampling Grid. From Section 4.4, the frequency sampling points $t_k = \frac{2\pi k}{\log N}$ have spacing $\Delta t = \frac{2\pi}{\log N} \rightarrow 0$ as $N \rightarrow \infty$. According to the Riemann-von Mangoldt zero density estimate $N(T) \sim \frac{T}{2\pi} \log T$, for the first J zeros $\gamma_1, \dots, \gamma_J$, when:

$$(67) \quad N > \exp\left(\frac{2\pi}{\min_{j \leq J} |\gamma_{j+1} - \gamma_j|}\right),$$

the grid can resolve all first J zeros (no omission, no aliasing).

8.2. FFT Phase Correspondence. At discrete points $t_k = \frac{2\pi k}{\log N}$:

$$(68) \quad D_N(t_k) \approx \sqrt{N} \cdot \text{FFT}[\Lambda(n)/\sqrt{n} \phi(\cdot)](k).$$

Phase error analysis:

$$(69) \quad \arg(D_N(t)) - \arg\left(-\frac{\zeta'(1/2 + it)}{\zeta(1/2 + it)}\right) = O\left(\frac{t^2}{\log N}\right) + O\left(\frac{1}{\sqrt{N}}\right)$$

For $t \in [0, T]$, selecting $N \sim T^2$ makes the error arbitrarily small.

8.3. Spectral Criteria for Zero Detection. At zeros $\rho = 1/2 + i\gamma$, $-\zeta'/\zeta$ has simple poles, and the phase undergoes π jumps.

In the FFT spectrum, this manifests as amplitude minima and phase discontinuities:

$$(70) \quad |\text{FFT}(k)| \approx \left| \frac{1}{\rho - (1/2 + it_k)} \right| \quad \text{near } t_k \approx \gamma,$$

forming Lorentzian-type peaks (or valleys, depending on normalization).

Theoretical Guarantee of Phase Jump Criterion: By Corollary 5.5, when t_k passes through γ_j from left to right, $\arg(D_N(t))$ undergoes a $+\pi$ jump (conversely $-\pi$). Detection of this jump is unaffected by amplitude normalization $|D_N(t_k)|^2/\|D_N\|_2^2$ and is a more robust numerical criterion.

Applicability: Theorem 5.2's L^2 error estimate is valid on $[0, T(N)]$ requiring $T(N) = o(\sqrt{N})$. Thus, to detect the first J zeros, we need $\gamma_J < \sqrt{N}$, i.e., $J \log J \lesssim \sqrt{N}$. This indicates N must grow quadratically with the number of zeros to be detected, consistent with the constructive protocol in Section 7.3.

9. CONCLUSION: MATHEMATICAL COMPLETENESS STATEMENT

Theorem 9.1 (NCDFT-RH Equivalence). *The following propositions are equivalent:*

- (1) **Riemann Hypothesis:** All non-trivial zeros satisfy $\Re(\rho) = 1/2$.
- (2) **Self-Duality:** $\mathcal{U}_\alpha^{(N)} = \mathbb{I}$ if and only if $\alpha = 1/2$.
- (3) **Spectral Compactification Uniqueness:** $\mathcal{H}_\alpha^{(N)} = 0$ (real part locked) if and only if $\alpha = 1/2$, and the spectral measure constructed through frequency indices t_k is compact (supported on the line $\Re(s) = 1/2$); when $\alpha \neq 1/2$, $\beta_n^{(N)} \rightarrow \infty$, and the spectral measure is non-compact in the complex plane (Carleman condition fails).
- (4) **Functorial Faithfulness:** $d_{\text{spec}}(\alpha) = 0$ if and only if $\alpha = 1/2$ (through correct limit ordering).
- (5) **FFT Convergence:** $\arg(\text{FFT}[\Lambda(n)/\sqrt{n}\phi(\cdot)])$ converges in the L^2 sense to $\arg\left(-\frac{\zeta'(1/2+it)}{\zeta(1/2+it)}\right)$.
- (6) **Sampling Completeness:** The discrete frequency grid $t_k = \frac{2\pi k}{\log N}$ can approximate all zero imaginary parts γ_j arbitrarily closely as $N \rightarrow \infty$, i.e., $\lim_{N \rightarrow \infty} \min_k |t_k - \gamma_j| = 0$ for all j (Lemma 5.3).

Sketch. (1) $\Rightarrow$ (2): By the Riemann functional equation, zeros on the critical line imply duality symmetry strictly holds at $\alpha = 1/2$, combined with Theorem 5.1 yielding $\mathcal{U}_{1/2} = \mathbb{I}$.

(2) $\Rightarrow$ (3): Self-duality forces $\mathcal{H}_{1/2} = 0$ (real part locked); if $\alpha \neq 1/2$ then $\mathcal{U}_\alpha \neq \mathbb{I}$, by Theorem 4.3 (all eigenvalues unbounded) and Theorem 4.6 (Jacobi coefficients diverging), non-compactness follows.

(3) $\Rightarrow$ (4): Spectral compactification ensures Weil trace formula convergence, functorial faithfulness holds; non-compact spectrum leads to deviation.

(4) $\Rightarrow$ (5): Functorial faithfulness ensures IDFT reconstruction matches explicit formula, FFT phase convergence follows.

(5)+(6) $\Rightarrow$ (1): By (5)'s FFT phase convergence, combined with (6)'s sampling completeness, for any γ_j , there exists subsequence $t_{k(N)} \rightarrow \gamma_j$ where $\arg(D_N(t_{k(N)}))$ exhibits π jumps. This implies zero real parts must be $1/2$, otherwise phase behavior would deviate from the pole characteristics of the logarithmic derivative. If a zero ρ_0 existed with $\Re(\rho_0) \neq 1/2$, its support would fall outside the critical line, contradicting the measure inclusion relation in Section 6.4. $\square$

Strict Derivation of RH:

From Theorems 3.3 and 4.5, when $\alpha = 1/2$, the scaled generator is strictly zero:

$$(71) \quad \mathcal{H}_{1/2}^{(N)} = 0$$

This corresponds to freezing of internal degrees of freedom—real part deviation is locked to zero, all spectral points are constrained to the line $\Re(s) = 1/2$. However, this only determines the real part coordinate of zeros. To recover the complete complex zeros $\rho_j = 1/2 + i\gamma_j$ (including imaginary part information), external frequency indices $t_k = \frac{2\pi k}{\log N}$ must be introduced through the frequency spectral measure $\nu_{1/2}^{(N)}$:

$$(72) \quad \nu_{1/2}^{(N)} := \frac{1}{\log N} \sum_{k=0}^{N-1} |D_N(t_k)|^2 \delta_{\frac{1}{2} + it_k}$$

The support of this measure is:

$$(73) \quad \text{supp}(\nu_{1/2}^{(N)}) = \left\{ \frac{1}{2} + it_k \mid k = 0, 1, \dots, N-1 \right\} \subset \left\{ s \in \mathbb{C} \mid \Re(s) = \frac{1}{2} \right\},$$

i.e., all mass is strictly distributed on the critical line $\Re(s) = 1/2$. By Theorem 6.1, as $N \rightarrow \infty$, this measure converges weakly to the empirical distribution of Riemann zeros:

$$(74) \quad \lim_{N \rightarrow \infty} \nu_{1/2}^{(N)} = \frac{1}{2\pi} \sum_{\rho} \delta_{\rho}.$$

By lower semicontinuity of support for weak convergence, the limit measure's support satisfies the inclusion relation:

$$(75) \quad \text{supp} \left(\sum_{\rho} \delta_{\rho} \right) \subseteq \overline{\bigcup_N \text{supp}(\nu_{1/2}^{(N)})} \subseteq \left\{ s \in \mathbb{C} \mid \Re(s) = \frac{1}{2} \right\}.$$

Therefore, all non-trivial zeros ρ must satisfy $\Re(\rho) = 1/2$. If a zero ρ_0 existed with $\Re(\rho_0) \neq 1/2$, the support of δ_{ρ_0} would fall outside the critical line, contradicting the above inclusion. $\square$

Key Distinction:

- Internal spectral measure $\mu_{1/2}^{(N)} = \delta_0$: Reflects $\mathcal{H}_{1/2}^{(N)} = 0$, only indicates real part locking (single-point compactification), contains no imaginary part information.
- Frequency spectral measure $\nu_{1/2}^{(N)}$: Reflects imaginary part information expanded through FFT frequencies t_k , supported on the critical line, converging to the zero distribution.

Together they constitute the complete spectral lifting: the “internal” part of $\mathcal{L}_{1/2}^{(N)}$ (from $\mathcal{H}$) locks the real part to $1/2$, while the “external” part (from $\text{diag}(t_k)$) expands the imaginary part to γ_j . Only when $\alpha = 1/2$ does this dual structure achieve strict self-dual matching. $\square$

Corollary: This framework provides a constructive proof pathway: approximating infinite-dimensional analytic objects (Riemann zeros) through finite-dimensional self-dual computations (standard DFT/FFT), with all steps satisfying Bishop’s constructivity requirements. The NCDFT framework reveals that the Riemann Hypothesis is essentially a duality condition: only when the parameter $\alpha = 1/2$ does the non-commutative discrete Fourier transform achieve strict self-duality, where the functorial faithfulness between arithmetic and spectrum reaches perfect matching, and the spectral measure compactifies from the non-compact extended state ($\alpha \neq 1/2$) to the discrete point state on the critical line ($\alpha = 1/2$).

APPENDIX A. FFT BUTTERFLY OPERATIONS AND DUALITY STRUCTURE

In Fast Fourier Transform (FFT), the butterfly operation is the core step implementing the divide-and-conquer algorithm. Its elegant algebraic structure not only embodies the symmetry of DFT but also has profound parallels with the duality $\mathcal{F}_{\alpha} \leftrightarrow \mathcal{F}_{1-\alpha}$ in the NCDFT framework. This appendix briefly presents the algebraic form of butterfly operations and explains their connection to the functional equation.

A.1. Butterfly Operation in Decimation-in-Time FFT. Consider the DFT of length N ($N = 2^m$):

$$(76) \quad X_k = \sum_{n=0}^{N-1} x_n \omega_N^{kn}, \quad \omega_N = e^{-2\pi i/N}.$$

Separate the input sequence into even and odd parts:

$$(77) \quad X_k = \sum_{m=0}^{N/2-1} x_{2m} \omega_N^{k(2m)} + \sum_{m=0}^{N/2-1} x_{2m+1} \omega_N^{k(2m+1)} = E_k + \omega_N^k O_k,$$

where E_k and O_k are DFTs of length $N/2$:

$$(78) \quad E_k = \sum_{m=0}^{N/2-1} x_{2m} \omega_{N/2}^{km}, \quad O_k = \sum_{m=0}^{N/2-1} x_{2m+1} \omega_{N/2}^{km}.$$

For k and $k + N/2$, using $\omega_N^{k+N/2} = -\omega_N^k$, we obtain the classic butterfly computation:

$$(79) \quad \begin{cases} X_k = E_k + \omega_N^k O_k, \\ X_{k+N/2} = E_k - \omega_N^k O_k. \end{cases}$$

This operation can be diagrammed as a butterfly structure, hence the name.

A.2. Butterfly Operation and Duality. The butterfly operation reveals the intrinsic symmetry of DFT: the even part E_k and odd part O_k combine through the twiddle factor ω_N^k , while the twiddle factor itself satisfies $\omega_N^{k+N/2} = -\omega_N^k$, which is precisely the discrete manifestation of the $s \leftrightarrow 1 - s$ symmetry in the functional equation. In the NCDFT framework, the duality $\mathcal{F}_\alpha \leftrightarrow \mathcal{F}_{1-\alpha}$ corresponds to the interchange of parameters α and $1 - \alpha$, while the even-odd decomposition and sign reversal in the butterfly operation precisely simulate this duality relationship.

Furthermore, viewing the NCDFT matrix $\mathcal{F}_\alpha^{(N)}$ as a DFT with internal phase $\exp(i(\alpha - 1/2) \text{Li}(x_n) \mathbf{H})$, its fast algorithm can similarly be decomposed into analogous butterfly structures, except the twiddle factors must be multiplied by corresponding diagonal matrices. This structure guarantees the computational efficiency of NCDFT while preserving the duality connection to the Riemann functional equation.

A.3. Butterfly Operation and Functional Equation. Combining the butterfly operation with the correspondence table from Section 4.2:

FFT Butterfly Operation	NCDFT Duality
Even-odd decomposition E_k, O_k	Parameter duality $\alpha \leftrightarrow 1 - \alpha$
Twiddle factors ω_N^k and $-\omega_N^k$	Phase factors $\exp(i(\alpha - 1/2) \text{Li}(x_n) \mathbf{H})$ and its conjugate
Symmetry of X_k and $X_{k+N/2}$	Self-dual condition $\mathcal{U}_{1/2} = \mathbb{I}$
Grid refinement $N \rightarrow \infty$	Sampling completeness $t_k \rightarrow \gamma_j$ (Section 4.4)

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